Chapter 3 / Stationarity and Power Spectral Density

Due: **1 week after Chapter 3 is completed**

Name: _____ Student ID: _____

Exercise 3.4

Required

Textbook p. 85

The random processes $X(t)$ and $Y(t)$ are independent and each is stationary. Prove that $Z(t) = X(t)Y(t)$ is also stationary.

Answer / Working

Chapter 3 / Stationarity and Power Spectral Density

Due: **1 week after Chapter 3 is completed**

Name: _____ Student ID: _____

Exercise 3.5

Required

Textbook p. 85

Let $X(t) = 10\sin(\omega_0 t + \Theta)$, where ω_0 is a deterministic constant and Θ is uniformly distributed on $[-\pi, \pi]$. A square-law device produces $Y(t) = X^2(t)$. Find:

(1) the mean of $Y(t)$;

(2) its autocorrelation function;

(3) whether $Y(t)$ is wide-sense stationary.

Answer / Working

Chapter 3 / Stationarity and Power Spectral Density

Due: **1 week after Chapter 3 is completed**

Name: _____ Student ID: _____

Exercise 3.9

Required

Textbook p. 85

The stationary random processes $X(t)$ and $Y(t)$ are statistically independent, have zero means, and have autocorrelation functions

$$R_X(\tau) = \mathrm{e}^{-|\tau|}, \qquad R_Y(\tau) = \cos(2\pi\tau).$$

(1) Find the autocorrelation of $Z(t) = X(t) + Y(t)$.

(2) Find the autocorrelation of $W(t) = X(t) - Y(t)$.

(3) Find the cross-correlation $R_{ZW}(\tau)$.

Answer / Working

Chapter 3 / Stationarity and Power Spectral Density

Due: **1 week after Chapter 3 is completed**

Name: _____ Student ID: _____

Exercise 3.14

Required

Textbook p. 86

Two zero-mean wide-sense stationary random processes $X(t)$ and $Y(t)$ have variances $\sigma_X^2 = 5$ and $\sigma_Y^2 = 10$. Can the following functions be their autocorrelation functions? Explain each answer. Here $u(\tau)$ denotes the unit step.

(1) $R_X(\tau) = 5u(\tau)\exp(-3\tau)$.

(2) $R_X(\tau) = 5\sin(5\tau)$.

(3) $R_Y(\tau) = 9/(1+2\tau^2)$.

(4) $R_Y(\tau) = -\cos(6\tau)\exp(-|\tau|)$.

(5) $R_X(\tau) = 5[\sin(3\tau)/(3\tau)]^2$.

(6) $R_Y(\tau) = 6 + 4\sin(10\tau)/(10\tau)$.

(7) $R_X(\tau) = 5\exp(-|\tau|)$.

(8) $R_Y(\tau) = 6 + 4\exp(-3\tau^2)$.

Removable singularities are understood by their limits.

Answer / Working

Chapter 3 / Stationarity and Power Spectral Density

Due: **1 week after Chapter 3 is completed**

Name: _____ Student ID: _____

Exercise 3.16

Required

Textbook p. 86

The processes $X(t)$ and $Y(t)$ are independent and each is stationary, with

$$R_X(\tau) = 2\mathrm{e}^{-|\tau|}\cos(\omega_0\tau), \qquad R_Y(\tau) = 9 + \exp(-3\tau^2).$$

Let $Z(t) = AX(t)Y(t)$, where the random variable A has mean 2 and variance 9, and A, $X(t)$ and $Y(t)$ are mutually independent. Find the mean, variance and autocorrelation of $Z(t)$.

Answer / Working

Chapter 3 / Stationarity and Power Spectral Density

Due: **1 week after Chapter 3 is completed**

Name: _____ Student ID: _____

Exercise 3.21

Required

Textbook p. 86–87

Which of the following functions can be power spectral densities of real random signals? Explain why or why not.

(1) $[\sin\omega/\omega]^2$.

(2) $\omega^2/(\omega^6+3\omega^2+3)$.

(3) $\omega^2/(\omega^4-1)-\delta(\omega)$.

(4) $\omega^4/(\mathrm{j}\omega^6+\omega^2+1)$.

(5) $|\omega|/(\omega^4+2\omega^2+1)$.

(6) $\exp[-(\omega-1)^2]$.

Answer / Working

Chapter 3 / Stationarity and Power Spectral Density

Due: **1 week after Chapter 3 is completed**

Name: _____ Student ID: _____

Exercise 3.25

Required

Textbook p. 87

Two random processes $X(t)$ and $Y(t)$ are jointly stationary, with cross-correlation

$$R_{XY}(\tau) = \begin{cases} 9\mathrm{e}^{-3\tau}, & \tau \geq 0, \\ 0, & \tau < 0. \end{cases}$$

Find the cross-power spectral densities $S_{XY}(\omega)$ and $S_{YX}(\omega)$.

Answer / Working

Chapter 3 / Stationarity and Power Spectral Density

Due: **1 week after Chapter 3 is completed**

Name: _____ Student ID: _____

Exercise 3.1

Optional

Textbook p. 85

A random voltage signal $U(t)$ is statistically independent at distinct times. Its one-dimensional distribution is Gaussian with mean 0 and variance 2.

(1) Find the density functions $f(u;t)$, $f(u_1,u_2;t_1,t_2)$ and $f(u_1,\ldots,u_k;t_1,\ldots,t_k)$, for arbitrary positive integer k.

(2) Determine the stationarity of $U(t)$.

Answer / Working

Chapter 3 / Stationarity and Power Spectral Density

Due: **1 week after Chapter 3 is completed**

Name: _____ Student ID: _____

Exercise 3.7

Optional

Textbook p. 85

The real random signal $Y(t)$ is wide-sense cyclostationary with cycle period 100. Its mean satisfies $m(10) = 20$, and its two-time correlation satisfies $R(5, 1) = 10$. Find:

(1) $\mathbb{E}[5Y(110)]$ and $\mathbb{E}[10Y(310) + 50]$;

(2) $\mathbb{E}[Y(105)Y(101)]$ and $\mathbb{E}[30Y(205)Y(201) + 200]$;

(3) $\mathbb{E}[10Y(305)Y(301) + 6Y(210) + 80]$.

Answer / Working

Chapter 3 / Stationarity and Power Spectral Density

Due: **1 week after Chapter 3 is completed**

Name: _____ Student ID: _____

Exercise 3.19

Optional

Textbook p. 86

Find the autocorrelation function and the mean-square value of a stationary signal $X(t)$ for each of the following power spectral densities:

(1) $S_X(\omega) = \omega^2/(\omega^4 + 3\omega^2 + 2)$.

(2)

$$S_X(\omega) = \begin{cases} 8\delta(\omega) + 20(1 - |\omega|/10), & |\omega| \le 10, \\ 0, & |\omega| > 10. \end{cases}$$

(3) $S_X(\omega) = p\cos^4(\omega\tau_0)$, where p is a constant.

Answer / Working